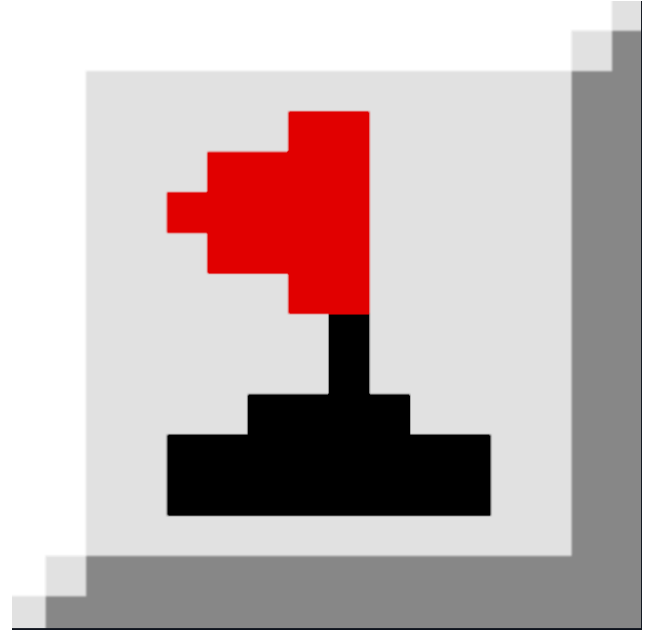


CTF: A Practical Introduction



WHOAMI

- Millie Solem (she/her)
- Robotics student @ University of Oslo
- Long-time CTF player and organizer
 - Organized **TG:Hack** since 2022
 - Norwegian ECSC team 2021+2022
 - Long-time member of Corax and bootplug
- Independent vulnerability researcher
 - Discovered 3 CVEs
 - Helped large organizations resolve security issues

What will be covered

- CTFs: what, where, how
- Learning to teach yourself
- DEMO: getting started with CryptoHack
- We solve some challenges together :))

What is a CTF?

Capture the Flag

A competition format where technical challenges are solved with a time limit. Each solved challenge yields a “flag”, which can be traded for points. The contestants with the most points at the end of a CTF win.

- Some CTFs **don't** have a time limit, these are called **Wargames**
- **FLAG{this_is_what_a_flag_might_look_like}**

Why play CTFs?

- Best way to learn practical skills
- Challenges can range in difficulty from beginner-level to expert
- Great community focused on sharing knowledge and improving
- Really, really fun

Finding CTFs

- CTF time: <https://ctftime.org/>
- Easy to set up a user and a team
- CTFs usually run in the weekend
- More fun with friends

Wargames

These run 24/7

- PicoCTF
- OverTheWire
- CryptoHack
- HackTheBox
- TryHackMe

DEMO

<https://cryptohack.org/>

A wargame focused on **Cryptography**

CYBERCHEF

<https://gchq.github.io/CyberChef/>

ASCII TABLE

Hex	Dec	Char	Hex	Dec	Char	Hex	Dec	Char	Hex	Dec	Char
0x00	0	NULL null	0x20	32	Space	0x40	64	@	0x60	96	`
0x01	1	SOH Start of heading	0x21	33	!	0x41	65	A	0x61	97	a
0x02	2	STX Start of text	0x22	34	"	0x42	66	B	0x62	98	b
0x03	3	ETX End of text	0x23	35	#	0x43	67	C	0x63	99	c
0x04	4	EOT End of transmission	0x24	36	\$	0x44	68	D	0x64	100	d
0x05	5	ENQ Enquiry	0x25	37	%	0x45	69	E	0x65	101	e
0x06	6	ACK Acknowledge	0x26	38	&	0x46	70	F	0x66	102	f
0x07	7	BELL Bell	0x27	39	'	0x47	71	G	0x67	103	g
0x08	8	BS Backspace	0x28	40	(	0x48	72	H	0x68	104	h
0x09	9	TAB Horizontal tab	0x29	41	)	0x49	73	I	0x69	105	i
0x0A	10	LF New line	0x2A	42	*	0x4A	74	J	0x6A	106	j
0x0B	11	VT Vertical tab	0x2B	43	+	0x4B	75	K	0x6B	107	k
0x0C	12	FF Form Feed	0x2C	44	,	0x4C	76	L	0x6C	108	l
0x0D	13	CR Carriage return	0x2D	45	-	0x4D	77	M	0x6D	109	m
0x0E	14	SO Shift out	0x2E	46	.	0x4E	78	N	0x6E	110	n
0x0F	15	SI Shift in	0x2F	47	/	0x4F	79	O	0x6F	111	o
0x10	16	DLE Data link escape	0x30	48	0	0x50	80	P	0x70	112	p
0x11	17	DC1 Device control 1	0x31	49	1	0x51	81	Q	0x71	113	q
0x12	18	DC2 Device control 2	0x32	50	2	0x52	82	R	0x72	114	r
0x13	19	DC3 Device control 3	0x33	51	3	0x53	83	S	0x73	115	s
0x14	20	DC4 Device control 4	0x34	52	4	0x54	84	T	0x74	116	t
0x15	21	NAK Negative ack	0x35	53	5	0x55	85	U	0x75	117	u
0x16	22	SYN Synchronous idle	0x36	54	6	0x56	86	V	0x76	118	v
0x17	23	ETB End transmission block	0x37	55	7	0x57	87	W	0x77	119	w
0x18	24	CAN Cancel	0x38	56	8	0x58	88	X	0x78	120	x
0x19	25	EM End of medium	0x39	57	9	0x59	89	Y	0x79	121	y
0x1A	26	SUB Substitute	0x3A	58	:	0x5A	90	Z	0x7A	122	z
0x1B	27	FSC Escape	0x3B	59	;	0x5B	91	[	0x7B	123	{
0x1C	28	FS File separator	0x3C	60	<	0x5C	92	\	0x7C	124	
0x1D	29	GS Group separator	0x3D	61	=	0x5D	93	]	0x7D	125	}
0x1E	30	RS Record separator	0x3E	62	>	0x5E	94	^	0x7E	126	~
0x1F	31	US Unit separator	0x3F	63	?	0x5F	95	_	0x7F	127	DEL

THANKS

My website: <https://solem.dev/>

Slides: <https://solem.dev/hackarinas.pdf>